

You are tasked with implementing a backend application for managing a shopping cart for an e-commerce system. Your target is to implement and deploy the first MVP API for adding items to a shopping cart on AWS. Below is an example of user story that you received from your team.

```
1  As a customer, I want to add items to my shopping cart
2  So that I can collect items I intend to purchase
3
4  Acceptance Criteria:
5  * I can add an item with specified quantity to my cart
6  * The cart should display the updated item count
7  * The cart should show the item details (product name, price, quantity)
8  * The cart should calculate and display the subtotal
```



You will be using Kiro for collaborative coding on this journey. You want to start slow and ensure certain software engineering practices/patterns will be applied.

## Prompt Example

You can use the prompt below as an initial conversation with Kiro CLI. Please take a few minutes to review and understand the prompt. Feel free to modify information as you see fit (such as programming language, solutions, architectural patterns, etc...).

```
1  I am a developer, I want to create a service to manage shopping cart API for e-commerce system.
2  I want to start with the "add items to cart" feature.
3
4  Below is the user story with acceptance criteria for this feature.
5  As a customer, I want to add items to my shopping cart
6  So that I can collect items I intend to purchase
7
8  Acceptance Criteria:
9  * I can add an item with specified quantity to my cart
10 * The cart should display the updated item count
11 * The cart should show the item details (product name, price, quantity)
12 * The cart should calculate and display the subtotal
13
14 The technical requirements are as follows
15 * Programming language is Python
16 * Follow Clean Architecture approach.
17 * Follow Test-Driven Development (TDD) approach. Enforce test first and follow red, green and refactor mechanism.
18 * Use AWS Cloud Development Kit (CDK) to implement infrastructure as code in Python
19 * Separate infrastructure and application folders. The infrastructure folder will be used to host the CDK code
20 * Use the following AWS solutions to build APIs: API Gateway, Lambda and DynamoDB
21
22 Please only do 1 feature at a time, start with add item to cart.
23 Also note the plan and update progress on scratchboard.md
```



After successfully building and deploying the application on AWS. You may try applying additional practices such as continuous integration and delivery, or add new features as well

## Useful Prompts

### Build and Deploy

In case you don't know the commands to build, run or deploy application, you can try with the following prompt

```
1  Could you help with building and deploying this application? Please also update progress on scratchboard.md.
```



### Run API Test

Similar to building and deploying, you can ask Kiro to run API test for you.

```
1  Could you help with running the API test? Please also update progress on scratchboard.md.
```



### Document Generation

Sometimes no document is generated. You can ask Kiro to generate a README file for your project as well.

```
1  Could you create a README file for this application? Please also update progress on scratchboard.md.
```



### Adding a New Feature

```
1  Based on current code, I would like to add an additional feature to shopping cart API.
2
3  Below is the user story for this feature
4  As a customer, I want to remove items from my shopping cart
5  So that I can adjust my intended purchases
6
7  Acceptance Criteria:
8  * I can remove a specific item completely from the cart
9  * The cart should update the total item count
10 * The cart should recalculate and display the updated subtotal
11
12 Remember to keep the technical requirements as previously mentioned.
13 Only do 1 feature at a time.
14
15 Also note the plan and update progress on scratchboard.md
```



## Tips and Tricks

- Start slow, focus on a specific task, understand the response from Kiro. Stop, verify and respond to polish the result if something doesn't feel right :).
- Provide detailed contexts to Kiro. The more you provide, the more accurate results you will get.
- Keep the plan and progress updated, ensure you tell Kiro to note down the plan and progress on scratchboard.md. This will help both you and Kiro track the progress and get more accurate results.

## Troubleshooting

If you encounter problems within the project such as misconfiguration, package or dependency mismatches, etc... , feel free to start over by deleting the code and type ``/clear`` on CLI to restart the session.

Note that if you get an error after deploying the application to AWS, you may need to delete the CloudFormation stack or apply a terraform plan to start over again.

## Raise your hand to get the prize!

Please raise your hand to get the prize!!! 🙋🎁

Thank you for completing this workshop!

## Clean up

Delete all resources created by this CDK Stack.

```
1  Delete all resources which created by this CDK Stack in my AWS Account
```

